Design and Control of a Reaction Wheel Testbed for Autonomous Tracking

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***Abstract*—Reaction wheels are commonly used in satellites to control orientation without external forces. This project presents the design, construction, and testing of a reaction wheel testbed capable of stabilizing a rotating platform and autonomously tracking a colored object. The system demonstrates conservation of angular momentum through controlled wheel motion and integrates computer vision with feedback control. Experimental results show effective disturbance rejection and successful target tracking, with minor limitations due to control tuning and sensor delays.**

I. Introduction

Reaction wheels are used by real satellites to change their orientation in space without external thrusters, and the objective of this project was to replicate this function. A free-spinning reaction wheel testbed was built so the angular speed and direction of the platform could be controlled. By doing this, the wheel's inertia and the conservation of angular momentum are used to intentionally rotate the whole platform. To demonstrate how this system works, a camera was mounted on the testbed. This allows a green object to be automatically tracked by spinning the platform to keep the color in constant view.

II. System Design

A. Mechanical Components

1) Base Design: Grace Ige: Designed the base for the reaction wheel testbed to keep the system stable and prevent tipping. The size and weight of the base were chosen by calculating the center of gravity and making sure the gravitational torque stayed within the base area. The part was 3D printed at Texas Inventionworks using the same infill density as other lab materials for consistency. The design includes a safety factor of 1.5 to handle expected loads and uses a friction fit to securely hold the REV Robotics UltraHex shaft.

2) Motor Support Design: Jinhun Heo: Designed the motor support using OnShape. Designed to attach to the testbed platform and the bearing at the bottom with screws and nuts (4 total). The actual motor will be fastened to the ceiling of the motor support with screws and nuts as well (6 total). The reason for this design choice was to utilize the screw holes, which would be much more reliable than a friction fit. The motor support is a cylindrical shape that is hollow on the inside for the motor to go in, and it uses four pillars instead of walls for wires to be easily accessed.

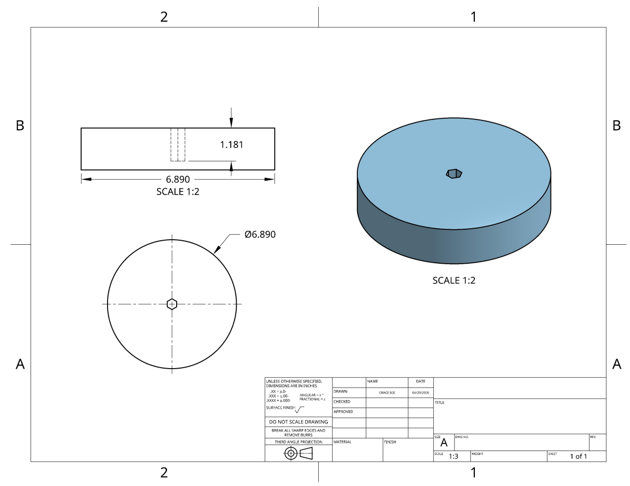

Fig. 1. Base design for the reaction wheel testbed

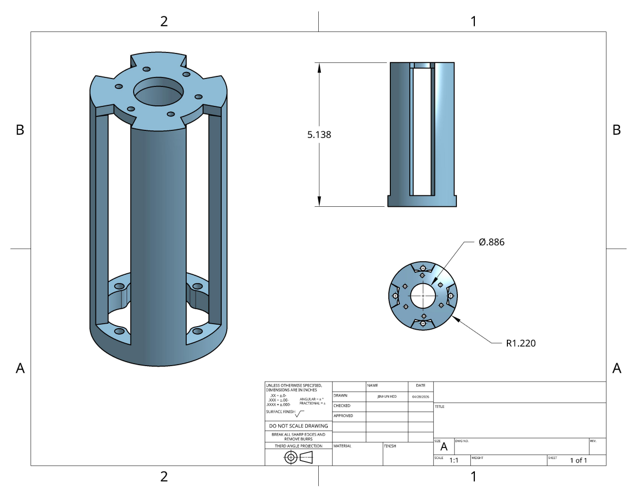

Fig. 2. Motor support structure with pillar-based design

3) Bearing Block Design: Haashir Azimi: A custom bearing block (Fig. 3) connects the testbed and the motor mount to the central hex rod. The base has four Ø0.169-inch clearance holes so it can be bolted securely to the motor mount. To avoid hitting other screws on the testbed, the base was designed with two flat edges.

Inside the tube, the holes are sized for a tight press fit around the bearings. To keep the assembly stable and prevent wobbling while spinning, the two bearings are spaced apart by three times their own height. The part was 3D printed on a Bambu P1P using PETG filament, which is both strong and affordable. For maximum durability without wasting material, it was printed with 5 outer shells and a 35

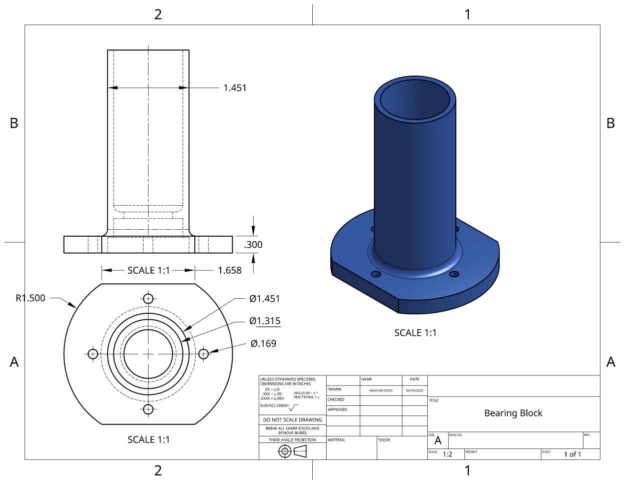

Fig. 3. Custom bearing block with press-fit design

4) Reaction Wheel Design: Adi Pullabhotla: The reaction wheel is spun by the motor and in turn the testbed platform spins the opposite direction to conserve angular momentum. The main objective of the design was to have enough inertia to prevent saturation. The wheel has a diameter of 240mm and thickness of 20mm. The wheel has a spoked design to create a hoop-like shape which increases moment of inertia. The further increase inertia, the wheel has 16 M5 and M6 clearance holes which alternate around the circumference of the wheel. This gives it a minimum mass of 195g and maximum mass of 432g. With the maximum mass the wheel has an average saturation time of 16.92 seconds. The wheel attaches directly on top of the motor with screws. There is a hex-shaft, however, this was omitted from the assembly to provide more stability while the wheel was spinning. Removing the motor hex-shaft also improves the entire system's tipping performance since it decreases the height and brings the center of mass lower.

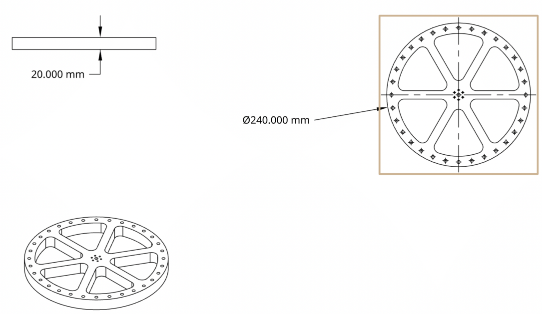

Fig. 4. Reaction wheel with adjustable mass distribution

III. System Integration

A. Assembly

The assembly process involved several design revisions to improve fit, stability, and ease of construction. The base was redesigned to better support the system and improve stability by adjusting its dimensions and shaft hole for a stronger friction fit . The motor support was modified to include four vertical pillars and a top ceiling, providing both lateral and vertical support; however, further adjustments are needed due to changes in motor height and gear ratio.

The bearing block was updated by refining tolerances and adding shaft collars to properly secure the bearing and prevent movement . The reaction wheel design was also revised by adding mounting holes for direct attachment to the motor and increasing the number of outer holes to allow additional mass, improving rotational inertia .

To improve assembly reliability, the team transitioned from tapped holes in plastic to clearance holes with nuts and bolts, reducing the risk of material failure . Additional holes were added to the testbed to allow more flexible fastening. Tolerances were adjusted across components, including tightening the bearing fit and increasing clearance for the motor housing to ensure proper assembly.

B. Hardware Integration

The assembled system integrates mechanical components with electronic control. The Raspberry Pi interfaces with:

- IMU for orientation data
- Motor driver via PWM signals
- Encoder for feedback control

IV. Wiring

The testbed's wiring was designed to keep high-power parts separate from sensitive electronics (Fig. 5). Power comes from a main battery. For safety, a fuse is placed right after the positive battery terminal. Being the first component in the circuit ensures the whole system is protected from short circuits or overheating. A main power switch is placed right after the fuse, allowing the entire system to be safely turned off at once.

To make assembly easier and work with available supplies, only two wire gauges were used. The main power lines, the motor connections, and the 5V power for the Raspberry Pi draw high continuous current (between 5.1 A and 13.6 A). To handle this safely, thick 18 AWG wire was used for all main power connections. On the other hand, the logic connections—such as the IMU sensor data, the motor control signals, and the encoder wires—draw very little current (less than 0.1 A). Therefore, thinner 24 AWG wire was used for these signals, which provides a good connection while keeping the wiring organized and compact.

The Raspberry Pi serves as the main computer. Tilt and rotation data are collected by an Inertial Measurement Unit (IMU) using a 3.3V I2C connection. To spin the reaction wheel, the Raspberry Pi sends PWM and direction signals to the H-Bridge motor driver. Finally, the motor's built-in encoder sends speed and position data back to the Raspberry Pi, allowing the system to precisely control the wheel's speed for stabilization.

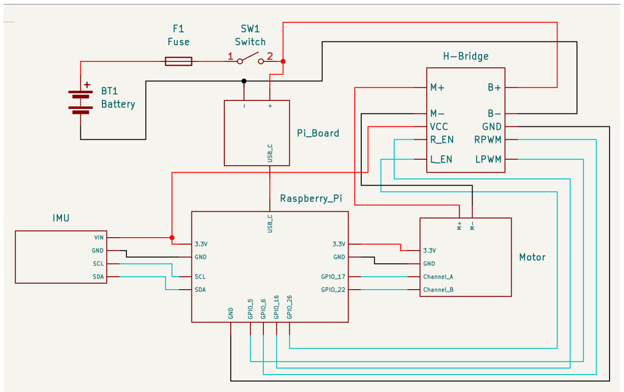

Fig. 5. Electrical wiring diagram of the system

V. Results

A. Mechanical Performance

First, the masses of the system components were measured. The base platform weighs 0.6 kg. Because the flywheel is modular, its mass can range from a minimum of 0.195 kg to a maximum of 0.432 kg. At its heaviest, the entire system weighs 1.465 kg. This means the flywheel makes up between 15.8% and 29.5% of the total system weight, depending on the configuration.

Next, the system was tested to see how freely the platform spins, ensuring the bearing block functioned correctly. Across three trials, the average spin-down time was 12.72 seconds (with a standard deviation of 0.39 seconds), demonstrating very little friction on the hex rod. The platform's stability was also tested, resulting in an average tipping angle of 42.1° (standard deviation of 0.6°).

Finally, the motor's saturation time was measured. This is the time it takes for the motor to reach its maximum speed, at which point it can no longer apply turning force to the platform. Over three trials, the average saturation time was 16.92 seconds, easily exceeding the 10-second design requirement. This time could be increased by adding additional nuts to the edge of the flywheel or making the wheel wider. However, this modification was avoided. While a heavier wheel takes longer to reach its maximum speed, it is also much slower to react, which would make the control system sluggish during the tracking phase.

B. Disturbance Rejection

To test the system's stability, the platform was manually pushed while it was programmed to stay still (0.0 RPM). As shown in (Fig. 7), these pushes caused sudden spikes in the platform's speed, reaching roughly -100 RPM and +40 RPM. In response, the PID controller immediately made the reaction wheel spin to counter the movement, briefly pushing the motor

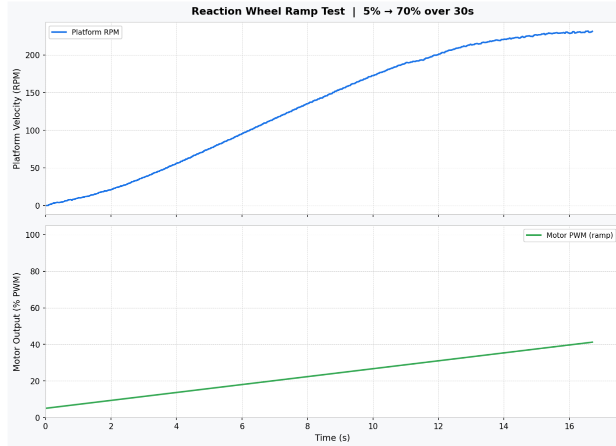

Fig. 6. Mechanical performance results

to its 100% maximum power. The system successfully handled these disturbances, bringing the platform back to a complete stop within one to two seconds. This proves the reaction wheel was strong enough to stabilize the testbed.

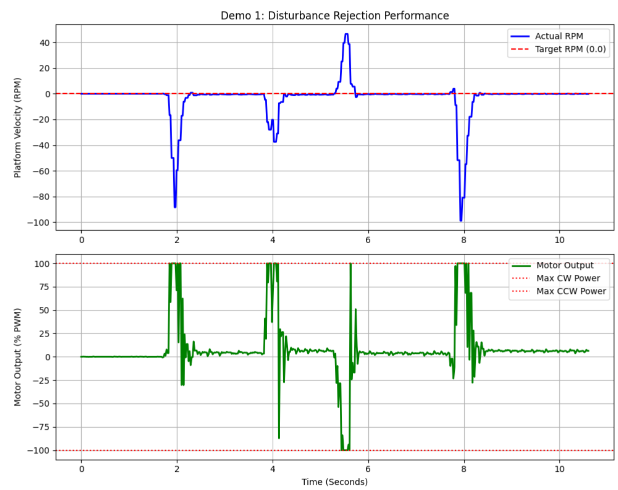

Fig. 7. Disturbance rejection performance

C. Autonomous Tracking

The final goal was to automatically track a green object by keeping it in the center of the camera's view (Fig. 8). The vision code successfully found the target and calculated the error by measuring how many pixels the green object was from the center of the camera frame. A two-part control system was used, as shown in (Fig. 9). The first part used the camera's pixel error to calculate a target speed for the platform. The second part used the IMU sensor to adjust the motor's power to match that target speed. The data shows that the platform's actual speed successfully matched the speed requested by the camera, allowing it to follow the target. However, the motor output graph shows rapid, small spikes in power. This caused the platform to shake or jitter slightly while tracking, likely due to a delay in the camera feed or imperfect tuning of the PID controller. Because of time constraints, this shaking was not completely smoothed out, but the overall system remained stable and worked as intended. .

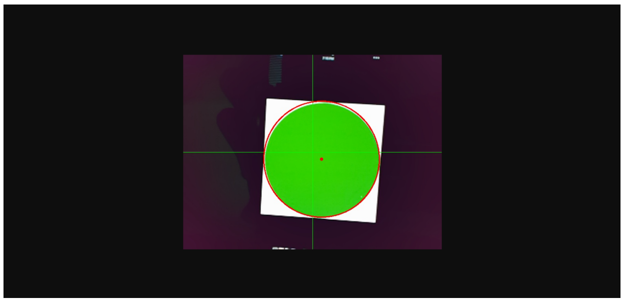

Fig. 8. Camera view during tracking

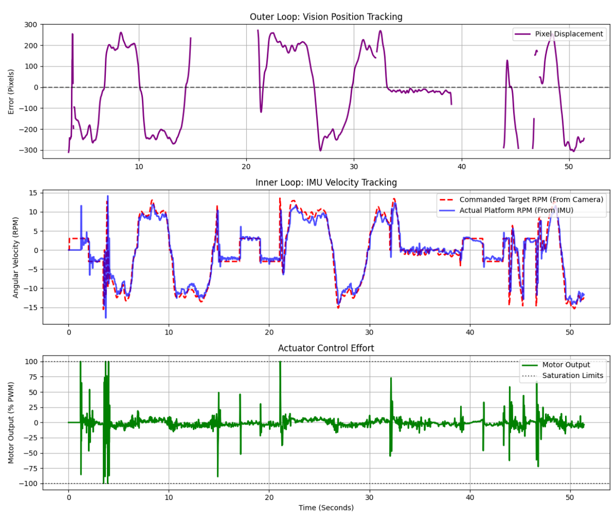

Fig. 9. Control response during tracking

VI. Future Work

While the reaction wheel testbed successfully demonstrated basic stabilization and color tracking, several improvements could be made. First, the rapid shaking observed during the tracking tests must be fixed. Future work will focus on filtering the IMU sensor data and better tuning the PID controller to smooth out the motor's power and stop the physical jitter.

Additionally, the camera tracking system can be upgraded. Instead of just looking for a specific color, a machine learning model like YOLO or MobileNet-SSD could be trained to track specific objects, like a tennis ball, even if the lighting changes. Because these advanced models require more computing power and can cause lag, this upgrade would likely require replacing the Raspberry Pi with a stronger computer, such as an NVIDIA Jetson Nano. To make the code easier to organize and manage, the entire software system could also be moved to ROS 2 (Robot Operating System).

Finally, the physical design of the flywheel can be improved. Although the current wheel meets the minimum saturation time, future designs will test moving more weight to the outside edge of the wheel. The goal is to increase the wheel's inertia, which gives the motor more time before it reaches its maximum speed, without making the system too slow to react to moving targets.

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